



◦ CSE 1113

Computer Programming

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Acknowledgement

Thanks to the authors of all the books and online tutorials used in this slide.



❑ TEXT BOOK:

There is no required text book

❑ Reference Books:

- Programming in ANCI C
By: E Balaguruswamy

Class Formats

- Lecture on grammars of C programming language
 - PowerPoint slides
 - We will have lots of questions and discussion
 - Lecture
- Practice in the lab
 - CSE 1114

Important Notice

- Academic Dishonesty
 - You must not turn in homework that is not yours.
 - You must prevent your work from being stolen.
- *Academic dishonesty or bad class behavior will be seriously penalized*



Lecture 1

Computer System and Programming

Computer?

- A programmable machine that
 - Receives input
 - Stores and manipulates data
 - Provides output in a useful format

Parts of a Computer



Webcamera



Speaker



Monitor



Speaker



Key-board



Mouse

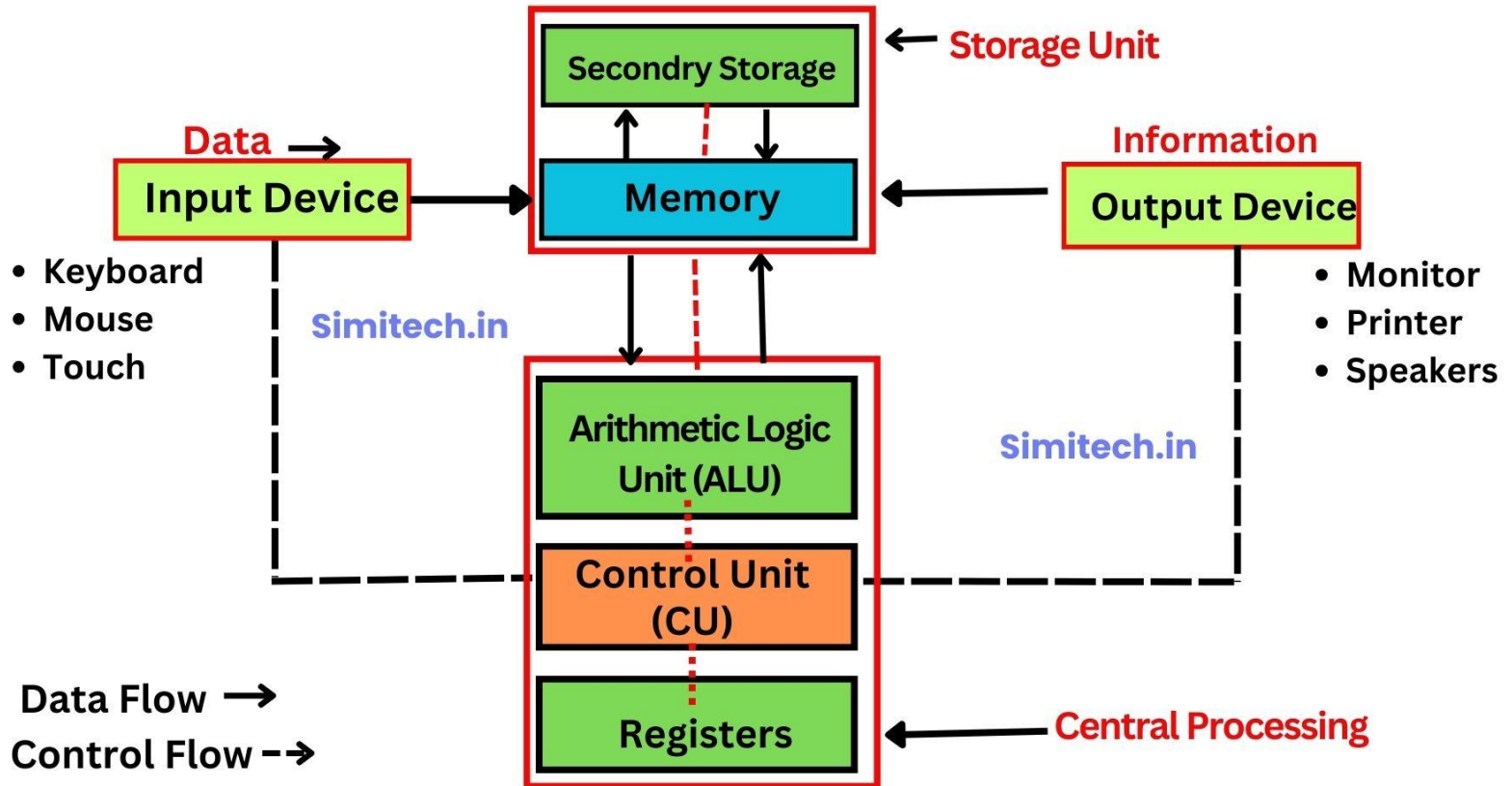


Printer



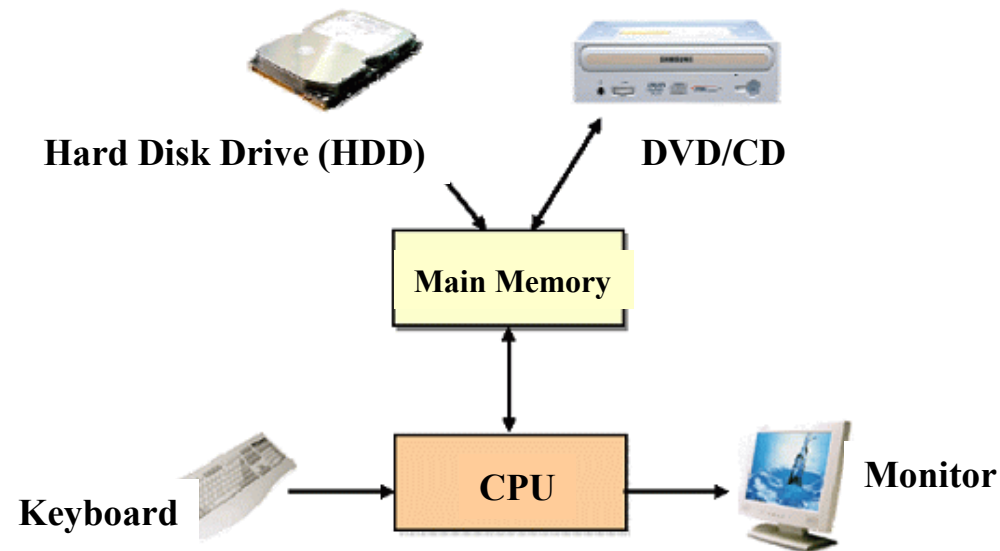
C. P. U

Block Diagram Of Computer



Computer System

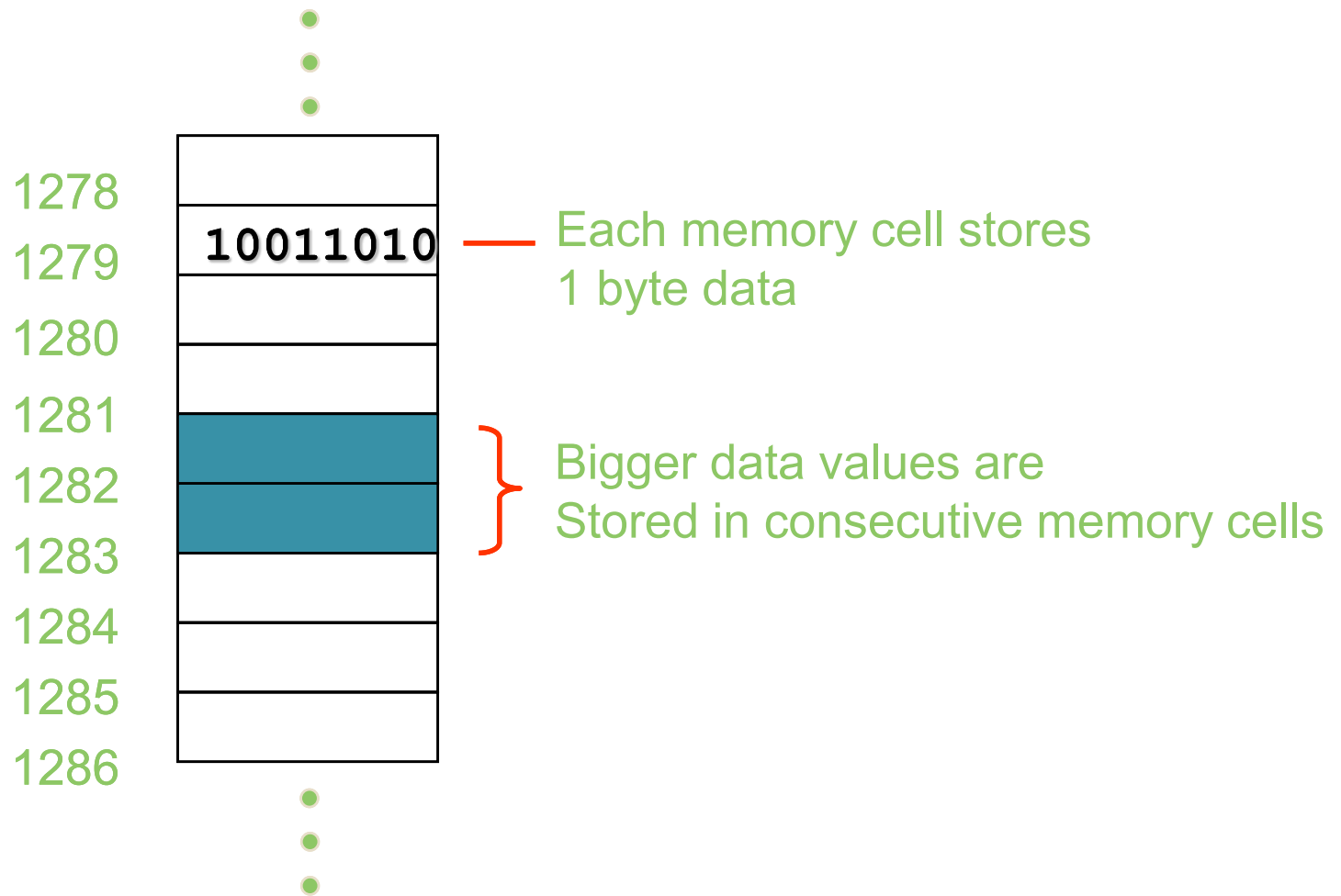
- Computer System
 - Hardware + Software
- Computer Hardware



Computer Hardware

- **CPU (Central Processing Unit)**
 - Processing program instructions (one by one)
 - Basic program instructions : add/subtract/multiply/div, read/write, jump, test
 - **Cache** : duplicating original data stored in slow storage into faster storage
- **Main Memory (e.g. RAM)**
 - **Volatile** : when power turned off, data in the memory will be erased
 - Storing program and data
 - Fast, small, and expensive
- **Secondary Memory (e.g. HDD, CD/DVD, ...)**
 - **Non volatile**
 - Relatively slow, large, and cheap
- **I/O(Input/ Output) Device**
 - Help interaction between computer and human beings.
 - Keyboard, mouse, monitor, etc

Memory and Data

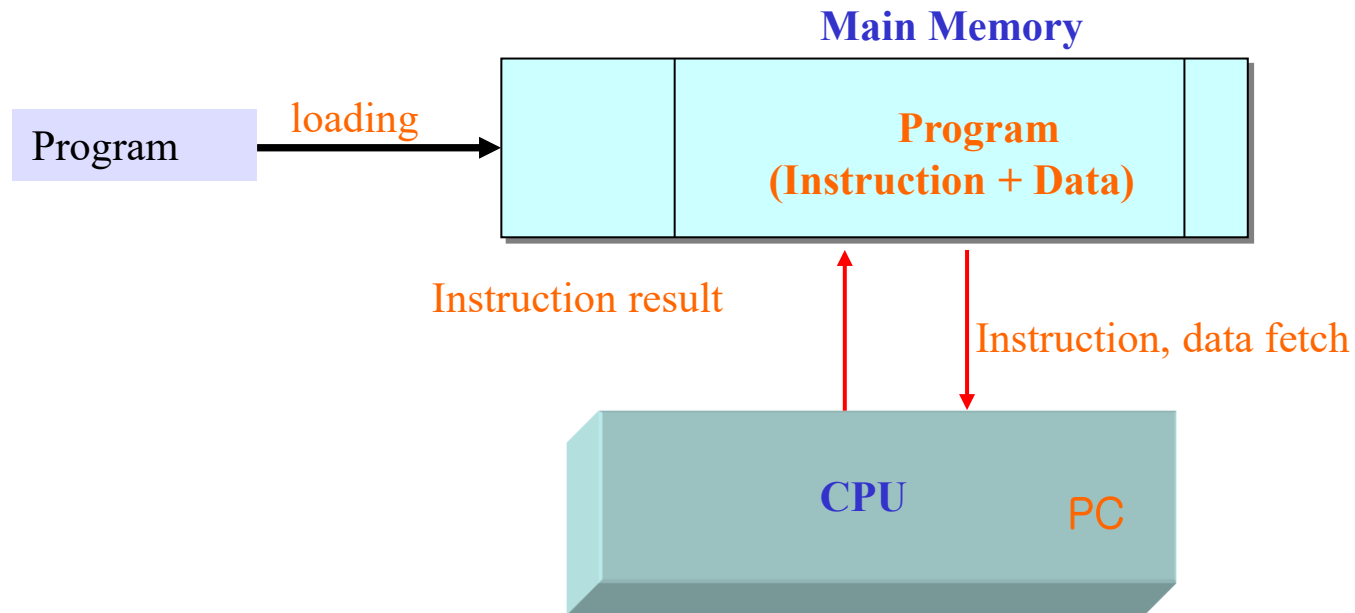


Units of Computer Memory Measurement

1.	1 bit	Binary digit
2.	8 bits	1 Byte
3.	1024 byte	1 KB (Kilo Byte)
4.	1024 KB	1 MB (Mega Byte)
5.	1024 MB	1 GB (Giga Byte)
6.	1024 GB	1 TB (Tera Byte)
7.	1024 TB	1 Peta Byte
8.	1024 PB	1 Exa Byte
9.	1024 EB	1 Zetta Byte
10.	1024 ZB	1 Yotta Byte
11.	1024 YB	1 Bronto Byte
12.	1024 BB	1 Geop Byte

Program Execution

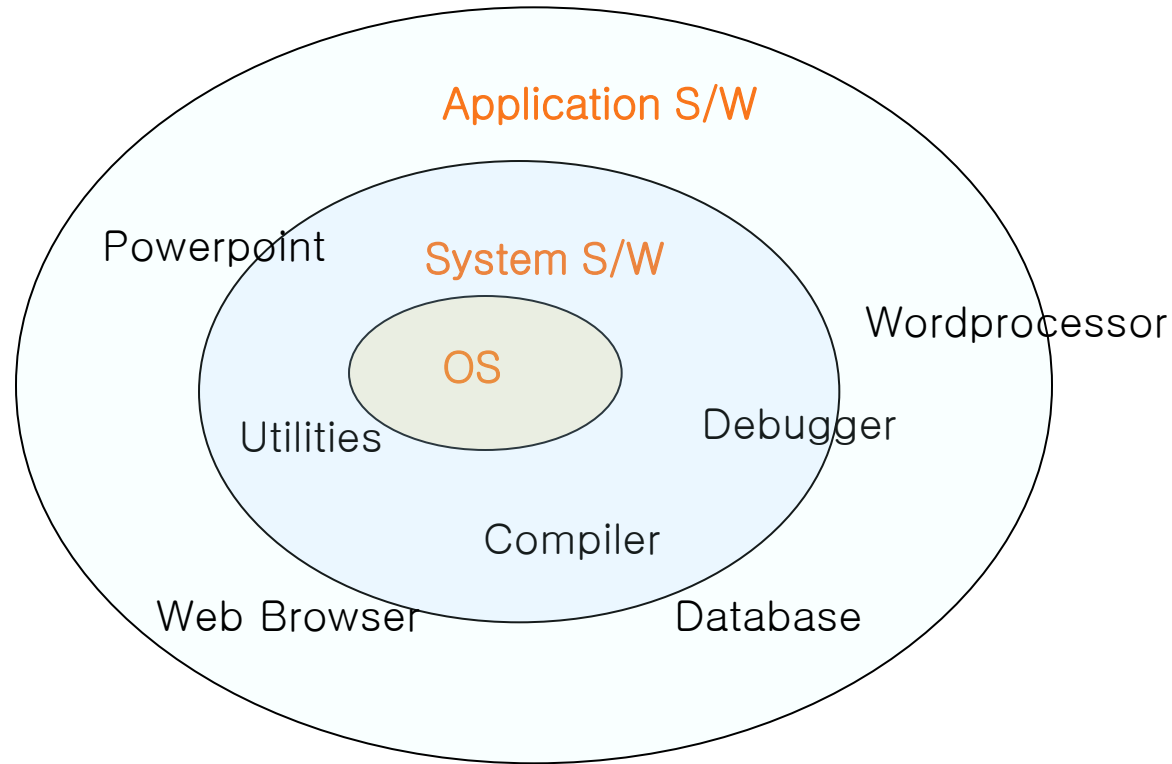
- Von Neumann architecture



Software

- System Software
 - Efficient management of computer system and resources
 - Operating System, Compiler, Debugger
- Application Software
 - All kinds of software other than system software
 - Word processor, spreadsheet(excel), graphics SW, artificial intelligence SW, Game SW, Statistics SW, medical SW

Software Layers



Data Representation

- Binary number
 - Computer uses binary number
 - 1 bit can represent 0 or 1
 - N bit number can represent up to 2^N

1 bit	2 bit	3 bit	4 bit
0	00	000	0000
1	01	001	0001
	10	010	0010
	11	011	0011
		100	0100
		101	0101
		110	0110
		111	0111
			1000
			1001
			1010
			1011
			1100
			1101
			1110
			1111

Binary number, decimal number

- Decimal number

- Use 0 - 9

- $182 = 1 \times 10^2 + 8 \times 10^1 + 2 \times 10^0$

$$= 1 \times 100 + 8 \times 10 + 2 \times 1$$

- Binary number

- Use 0 and 1

- $1101_2 = 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$

$$= 1 \times 8 + 1 \times 4 + 0 \times 2 + 1 \times 1$$

$$= 13$$

- Octal number

- Hexadecimal number

Programming Language

- Language for programming computer processing
 - Machine readable language designed to express computations that can be performed by a computer
 - Specify behavior of machine, express algorithms
 - Human-Computer Communications
- Machine language
 - Binary code
 - 1001 0001 store value at address 0001 into accumulator
 - 1100 0010 add value at address 0010 into accumulator
- Assembly Language
 - Symbolization of machine language binary code
 - LOAD Y
 - ADD Z

Programming Language

- High level language
 - Easy to use (read and write) , human friendly
 - Programmer does not need to know details of machine control.
 - More portable (machine independent)

$$X = Y + Z$$

- example
 - FORTRAN, COBOL, BASIC, C, C++, Java, Python

High Level Programming Language

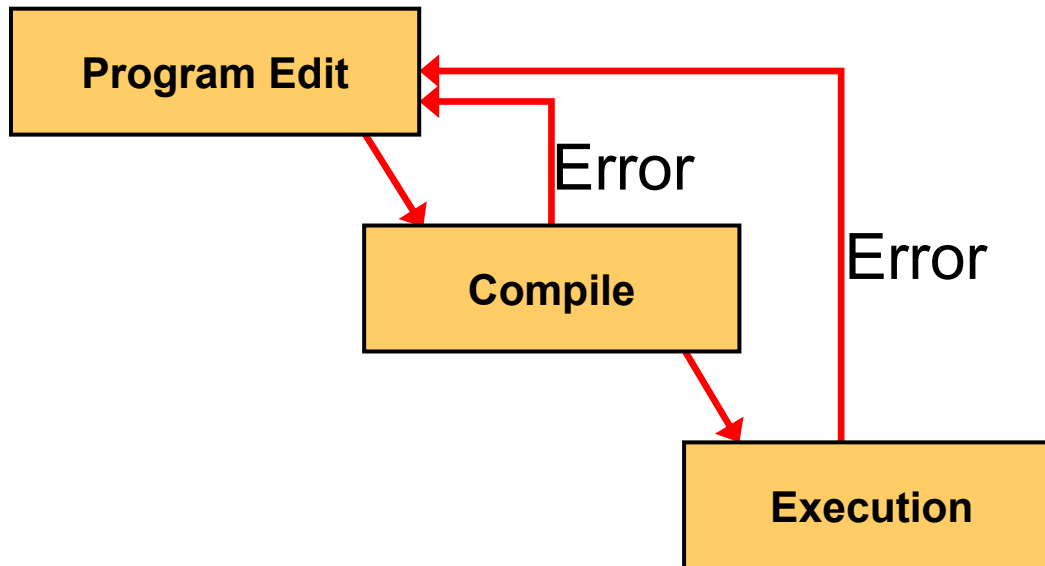
- **FORTRAN(FORmula TRANslation)**
 - Created in 1957 by John Backus (IBM)
 - General purpose PL especially suited For scientific computation
- **COBOL(COMmon Business Oriented Language)**
 - Created in early 1960s
 - Primarily used for business, finance in companies and govenment
- **BASIC(Beginner's All-purpose Symbolic Instruction Code)**
 - Easy to learn and use for beginners

High Level Programming Language

- **C**
 - Made by Dennis Ritchie (AT&T Bell Lab)
 - made for developing UNIX OS (1970s)
 - High level language with low level language properties (pointers,...)
- **C++**
 - Made by B. Stroustrup (AT&T Bell Lab)
 - OOPL(Object Oriented Programming Language) extending C
- **Java**
 - Made by James Gosling (Sun Microsystems, 1990s)
 - Platform independent OOPL
- **Python**
 - Made by Created by Guido van Rossum in 1991
 - it has become one of the most popular programming languages globally.
 - Platform independent OOPL

Programming and Execution

- **Programming Tool**
 - Editor, Compiler, Interpreter, Debugger, and etc
 - Integrated Development Environment (**IDE**)



Error

- **compile-time error**
 - Error occurring during compilation
 - Grammar check
 - Cannot execute if there is compile error
- **logical error**
 - Grammar is OK but logical error
- **run-time error**
 - Abnormal termination owing to unexpected reasons during program execution
 - Ex → divided by zero, illegal memory access

Debugging

- **debugging**
 - Bug : program error
 - Debugging : bug correction

Compiler / Interpreter

- Compiler
 - Convert high level language to low level language
(occur at compile-time)
- Interpreter
 - Compile and execute the program line by line
(occur at run-time)
- Comparison?



Part II

Program Creating, Compiling, Debugging and Running

A Simple C Program

```
1  /* A first program in C */
2  #include <stdio.h>
3
4  int main()
5  {
6      printf( "Welcome to DUET!\n" );
7
8      return 0;
9 }
```

Welcome to C!

- Comments
 - Text surrounded by `/*` and `*/` is ignored by computer
 - Used to describe program
- `#include <stdio.h>`
 - Preprocessor directive
 - Tells computer to load contents of a certain file
 - `<stdio.h>` allows standard input/output operations

A Simple C Program, Cont.

- `int main()`
 - C programs contain one or more functions, exactly one of which must be `main`
 - Parenthesis `()` used to indicate a function
 - `int` means that `main` "returns" an integer value
 - Braces `{` and `}` indicate a block
 - The bodies of all functions must be contained in braces

A Simple C Program: Printing a Line of Text

- `printf( "Welcome to DUET!\n" );`
 - Instructs computer to perform an action
 - Specifically, prints the string of characters within quotes (" ")
 - Entire line called a statement
 - All statements must end with a semicolon (;)
 - Escape character (\)
 - Indicates that printf should do something out of the ordinary
 - \n is the newline character

A Simple C Program: Printing a Line of Text

- `return 0;`
 - A way to exit a function
 - `return 0`, in this case, means that the program terminated normally

Basic Structure of C Program

Documentation Section

Link Section

Definition Section

Global Declaration Section

main () Function Section

{

Declaration part

Executable part

}

Subprogram section

Function 1

Function 2

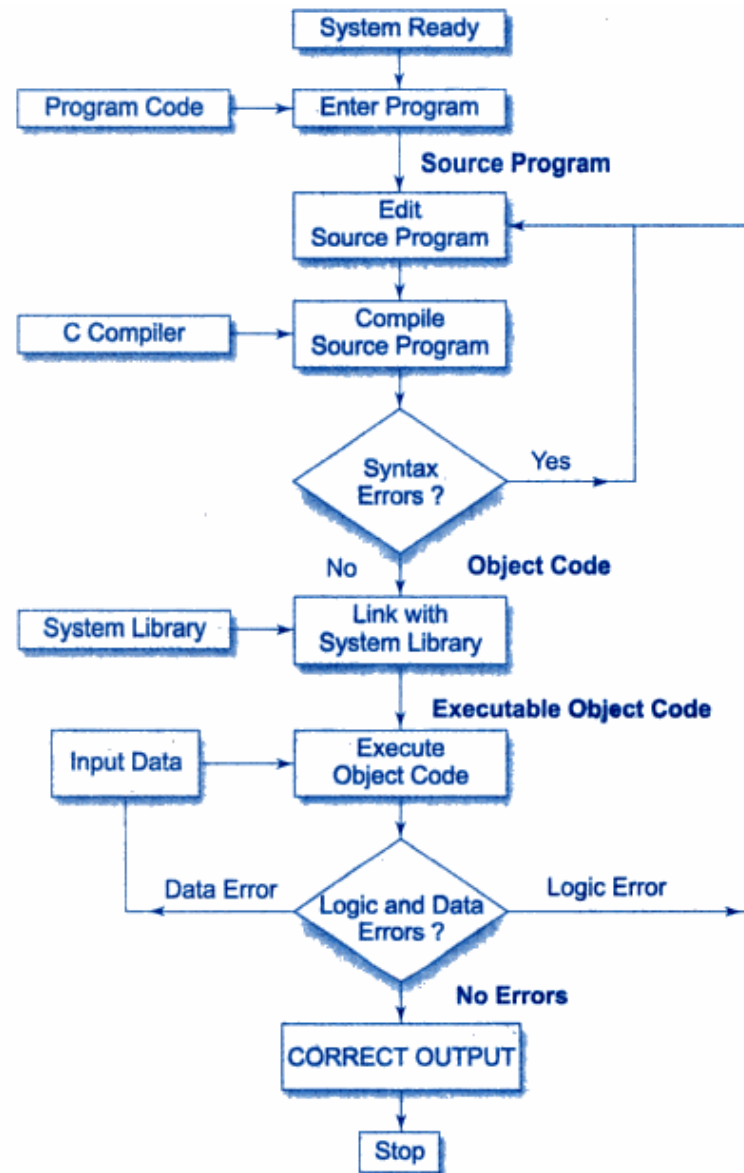
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Function n

(User-defined functions)

Executing a C Program



**Thank You
For
having Patience**



Best of Luck !